



**Mathematics Extension 1**  
**Year 11 Assessment Task 2**  
**Week 4 Term 3 2021**

**Name:** .....

**Teacher:** .....

**Thursday 5<sup>th</sup> August 2021**

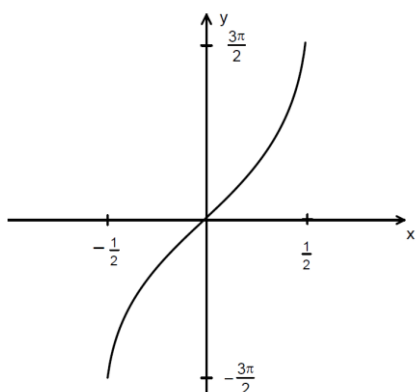
**General Instructions**

- All answers are to be written on your own lined paper so they can be scanned and uploaded as a single PDF file to your teacher's google classroom. Make sure that you write your name on each of the lined pages used.
- **Start each question on a new page.**
- Calculators approved by NESA may be used.
- Write using black pen.
- In Questions 1– 5, show relevant mathematical reasoning and/or calculations. Marks may be deducted for insufficient or illegible work.
- For any multiple-choice question, **write the letter** that corresponds to your answer on your writing paper.
- Time Allowed: **55 minutes inclusive of task downloading then answering on your own lined paper, scanning and uploading a single PDF file for your responses to your teacher's google classroom.**
- **Total marks – 32**

**Setter: YUS**

**Question 1** (6 marks)

- (a) Which of the alternatives could represent the inverse trigonometric function drawn below?



- A.  $y = \frac{1}{3} \sin^{-1} \frac{x}{2}$
- B.  $y = \frac{1}{3} \sin^{-1} 2x$
- C.  $y = 3 \sin^{-1} \frac{x}{2}$
- D.  $y = 3 \sin^{-1} 2x$

- (b) If  $\log_5 8 = a$ , prove that  $\log_{10} 2 = \frac{a}{a+3}$ . **2**

- (c) An inverted cone of radius  $r = 40$  cm and depth  $h = 70$  cm is filled with water. The water flows out of a hole in the apex of the cone at a constant rate of  $30\pi \text{ cm}^3/\text{minute}$ .

- (i) Show that the volume  $V$  of the cone is given by  $V = \frac{16\pi h^3}{147}$ . **1**

- (ii) Find the rate at which the water level is decreasing when the depth of water is 10cm. **2**

**End of Question 1**

**Question 2** (6 marks) Start a new page.

A baked cake cools at a rate proportional to the difference between the cake temperature ( $T$ ) and the room temperature ( $20^{\circ}\text{C}$ ). Initially the cake is  $80^{\circ}\text{C}$  and after 10 minutes the cake is  $50^{\circ}\text{C}$ .

- (i) Show that  $T = 20 + Ae^{kt}$  satisfies the cooling equation  $\frac{dT}{dt} = k(T - 20)$ . 1
- (ii) Show that  $A = 60$ . 1
- (iii) Find the exact value of  $k$  and hence find when the cake temperature reaches  $35^{\circ}\text{C}$ . 2
- (iv) Find the limiting temperature of the cake. 1
- (v) What is the rate of cooling after exactly 10 minutes? 1

**End of Question 2**

**Question 3** (7 marks) Start a new page.

- (a) A driver takes 3 hours to travel the distance between two points  $P$  and  $Q$  on a country road. At time  $t$  hours after passing  $P$ , the driver's speed  $v$  km/h is given by  $v = 70 + 40e^{-t}$ .
  - (i) Calculate the speed when the driver passes point  $Q$ , to the 2 significant figures. 1
  - (ii) Write the acceleration in terms of  $t$ . 1
  - (iii) Sketch the velocity-time curve and comment on the velocity for large values of  $t$ . 2
- (b) Find the value of  $\tan\left(\frac{7\pi}{12}\right)$  as a simplified surd with a rational denominator. 3

**End of Question 3**

**Question 4** (6 marks) Start a new page.

- (a) Using the  $t$ -substitutions, or otherwise, prove the identity

3

$$\frac{\tan 2\theta - \tan \theta}{\tan 2\theta + \cot \theta} = \tan^2 \theta$$

- (b)

$$\text{For the function } f(x) = \begin{cases} \frac{1}{x} & \text{for } 0 < x < 2 \\ \frac{1}{2} & \text{for } x = 2 \\ 1 - \frac{1}{4}x & \text{for } x > 2 \end{cases}$$

- (i) Show that the curve is continuous at  $x = 2$ .
- (ii) Determine whether the function is differentiable at  $x = 2$  and if so, state the value of  $f'(2)$ . Explain your reasoning carefully.

1

2

**End of Question 4**

**Question 5** (7 marks) Start a new page.

- (a) Determine where the curve  $y = \sin 5x + \sin x$  cuts the  $x$ -axis for  $0 \leq x \leq \frac{\pi}{2}$ .

2

- (b) Consider the function  $y = \tan^{-1}(\tan x)$ .

- (i) State its domain and whether it is even, odd or neither

2

- (ii) Simplify  $\tan^{-1}(\tan x)$  for  $-\frac{\pi}{2} < x < \frac{\pi}{2}$ .

1

- (iii) Use the above information and a table of values if necessary to sketch the function for  $-\frac{\pi}{2} \leq x \leq \frac{3\pi}{2}$ . You may assume  $\tan x = \tan(x + \pi)$ .

2

**End of paper**



Suggested Solution (s)

Comments

Question 1

a)  $-\frac{1}{2} \leq x \leq \frac{1}{2}$

$-1 \leq 2x \leq 1$

&  $-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$

$-\frac{\pi}{2} \leq \frac{y}{3} \leq \frac{\pi}{2}$

So  $\frac{y}{3} = \sin^{-1} 2x$

(ie)  $y = 3\sin^{-1} 2x$

b)  $\log_5 8 = a \Rightarrow \log_5 2^3 = a$

(ie)  $\log_5 2 = \frac{a}{3}$

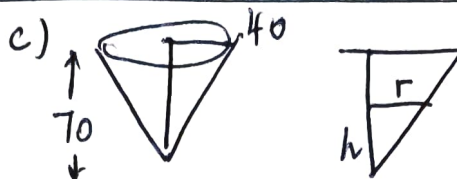
Now  $\log_{10} 2 = \frac{\log_5 2}{\log_5 10}$

$= \frac{\log_5 2}{\log_5 (2 \times 5)}$   
 $= \frac{\log_5 2}{\log_5 2 + \log_5 5}$  or  $\checkmark$

$\therefore = \frac{a/3}{a/3 + 1}$  using  $(*)$

$= \frac{a/3}{a/3 + 3/3}$

$= \frac{a}{a+3}$



i)  $V = \frac{1}{3} \pi r^2 h$  but  $\frac{r}{h} = \frac{40}{70}$

$r = \frac{4h}{7}$

$\therefore V = \frac{1}{3} \pi \left( \frac{4h}{7} \right)^2 h$   $\checkmark$   
 $= \frac{16\pi h^3}{147}$

ii) Now  $\frac{dV}{dh} = \frac{48\pi h^2}{147}$

$\frac{dh}{dt} = \frac{dh}{dV} \cdot \frac{dV}{dt}$

$= \frac{147}{48\pi h^2} \times -30\pi$   $\checkmark$

Now  $h = 10$

$\frac{dh}{dt} = \frac{147 \times -30\pi}{48\pi (10)^2}$

$= \frac{-4410}{4800}$   $\checkmark$

$= -0.91875 \text{ cm/min.}$

or  $-\frac{147}{160} \text{ cm/min.}$

$\therefore$  The depth of water is decreasing at  $0.92 \text{ cm/min.}$



Year 11

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SOLUTIONS

(2)

## Suggested Solution (s)

## Comments

Question 2

(i)  $T = 20 + Ae^{kt}$

$$\begin{aligned} \text{LHS} = \frac{dT}{dt} &= Ae^{kt} \times k \\ &= Ake^{kt} \end{aligned}$$

but  $Ae^{kt} = T - 20$  ✓

$$\therefore \frac{dT}{dt} = k(T - 20)$$

= RHS

(ii) When  $t = 0$   $T = 80$

ie)  $80 = 20 + Ae^0$  ✓

$A = 60$

(iii) When  $t = 10$ ,  $T = 50$

ie)  $50 = 20 + 60e^{10k}$

$\frac{30}{60} = e^{10k}$

$\ln \frac{1}{2} = 10k$

$-\ln 2 = 10k$

$\therefore k = -\frac{\ln 2}{10}$  ✓

When  $T = 35$ ,  $t = ?$

ie)  $35 = 20 + 60e^{kt}$  where  $k = \frac{-\ln 2}{10}$

$\frac{15}{60} = e^{kt}$

$\ln \frac{1}{4} = kt$

$t = \frac{\ln \frac{1}{4}}{k}$

$t = 20 \text{ min}$  ✓

iv) As  $t \rightarrow \infty$   $60e^{kt} \rightarrow 0$

$\therefore T \rightarrow 20$

The limiting temperature is  $20^\circ$  ✓

v)  $\frac{dT}{dt} = k(T - 20)$

$= k(50 - 20)$

where  $k = -\frac{\ln 2}{10}$

$= -2.079441$

The rate of cooling

is  $2.079^\circ\text{C/min}$  ✓



Year 11

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SOLUTIONS

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Suggested Solution (s)

Comments

### Question 3

a)  $V = 70 + 40e^{-t}$

i) For Q,  $t = 3$

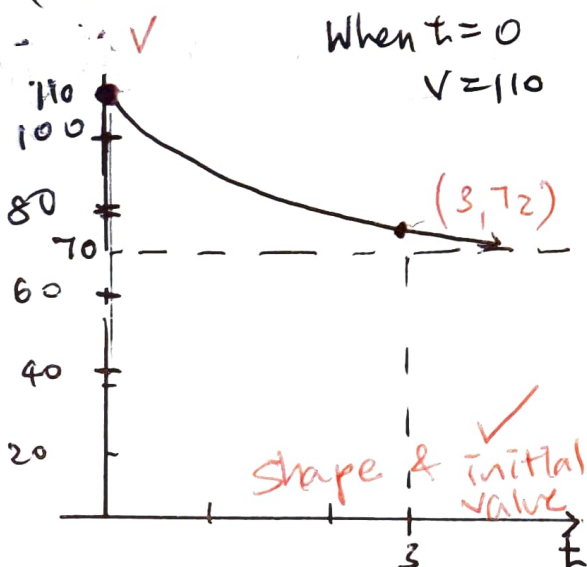
$\therefore V = 70 + 40e^{-3}$

$= 71.99$

$\approx 72 \text{ km/h}$  ✓

ii)  $a = -40e^{-t}$  ✓

(iii)



As  $t \rightarrow \infty$ ,  $40e^{-t} \rightarrow 0$

$\therefore V \rightarrow 70$

as  $t$  increases the velocity approaches  $70 \text{ km/h}$  ✓

(b)  $\tan \frac{7\pi}{12} = \tan \left( \frac{3\pi}{12} + \frac{4\pi}{12} \right)$

ie)  $\tan \left( \frac{\pi}{4} + \frac{\pi}{3} \right)$

$= \frac{\tan \frac{\pi}{4} + \tan \frac{\pi}{3}}{1 - \tan \frac{\pi}{4} \tan \frac{\pi}{3}}$  ✓

$= \frac{1 + \sqrt{3}}{1 - \sqrt{3}}$  ✓

$= \frac{(1 + \sqrt{3})(1 + \sqrt{3})}{(1 - \sqrt{3})(1 + \sqrt{3})}$

$= \frac{1 + 2\sqrt{3} + 3}{-2}$

$= -2 - \sqrt{3}$  ✓



## Suggested Solution (s)

## Comments

Question 4Method I

Let  $\tan \theta = t$

$$\text{LHS} = \frac{\frac{2t}{1-t^2} - t}{\frac{2t}{1-t^2} + \frac{1}{t}}$$

$$\text{RHS} = t^2$$

$$= \frac{2t - t(1-t^2)}{1-t^2} \div \frac{2t^2 + (1-t^2)}{t(1-t^2)} \quad \checkmark$$

$$= \frac{2t - t + t^3}{(1-t^2)} \times \frac{t(1-t^2)}{2t^2 + 1 - t^2} \quad \checkmark$$

$$= (t^3 + t) \times \frac{t}{t^2 + 1} \quad \checkmark$$

$$= t(t^2 + 1) \times \frac{t}{(t^2 + 1)} \quad \checkmark$$

$$= t^2$$

$$= \text{RHS}$$

Method II

$$\frac{\frac{2 \tan \theta}{1 - \tan^2 \theta} - \frac{\tan \theta}{1}}{\frac{2 \tan \theta}{1 - \tan^2 \theta} + \frac{1}{\tan \theta}}$$

$$= \frac{(\tan \theta + \tan^3 \theta) / (1 - \tan^2 \theta)}{(\tan^2 \theta + 1) / \tan \theta (1 - \tan^2 \theta)} \quad \checkmark$$

$$= \frac{\tan \theta (1 + \tan^2 \theta)}{(1 - \tan^2 \theta)} \times \frac{\tan (1 - \tan^2 \theta)}{(\tan^2 \theta + 1)} \quad \checkmark$$

$$= \tan^2 \theta$$



2021 Year 11 Mathematics Extension 1 Task 2 SOLUTIONS

Suggested Solution (s)

Comments

Question 4 (b)

(i) Given  $f(2) = \frac{1}{2}$ , we require  $\lim_{x \rightarrow 2^-} f(x) = f(2) = \lim_{x \rightarrow 2^+} f(x)$ .

Now,  $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} \frac{1}{x}$  or  $\frac{1}{2}$  &  $\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} (1 - \frac{1}{4}x)$   
 $= 1 - \frac{1}{4} \times 2$  or  $\frac{1}{2}$ .

As  $\lim_{x \rightarrow 2^-} f(x) = f(2)$  or  $\frac{1}{2} = \lim_{x \rightarrow 2^+} f(x)$ ,  $f$  is cti at  $x=2$

(ii) We require  $\lim_{h \rightarrow 0^-} \frac{f(2+h) - f(2)}{h}$  &  $\lim_{h \rightarrow 0^+} \frac{f(2+h) - f(2)}{h}$  to

exist with  $\lim_{h \rightarrow 0^-} \frac{f(2+h) - f(2)}{h} = \lim_{h \rightarrow 0^+} \frac{f(2+h) - f(2)}{h}$

Now,  $\lim_{h \rightarrow 0^-} \frac{f(2+h) - f(2)}{h}$

$= \lim_{h \rightarrow 0^-} \frac{\frac{1}{2+h} - \frac{1}{2}}{h}$

$= \lim_{h \rightarrow 0^-} \left( \frac{2 - (2+h)}{2(2+h)} \times \frac{1}{h} \right)$

$= \lim_{h \rightarrow 0^-} \frac{-h}{2h(2+h)}$

$= \lim_{h \rightarrow 0^-} \frac{-1}{2(2+h)}$

$= -\frac{1}{4}$

&  $\lim_{h \rightarrow 0^+} \frac{f(2+h) - f(2)}{h}$

$= \lim_{h \rightarrow 0^+} \frac{1 - \frac{1}{4}(2+h) - \frac{1}{2}}{h}$

$= \lim_{h \rightarrow 0^+} \frac{1 - \frac{1}{2} - \frac{1}{4}h - \frac{1}{2}}{h}$

$= \lim_{h \rightarrow 0^+} \frac{-\frac{1}{4}h}{h}$

$= \lim_{h \rightarrow 0^+} -\frac{1}{4}$

$= -\frac{1}{4}$

As  $\lim_{h \rightarrow 0^-} \frac{f(2+h) - f(2)}{h} = \lim_{h \rightarrow 0^+} \frac{f(2+h) - f(2)}{h}$ , conclude

that the function is differentiable at  $x=2$  with  $f'(2) = -\frac{1}{4}$ .

2 marks  
for  
argument.



## Suggested Solution (s)

## Comments

Question 5

a)  $y = \sin 5x + \sin x$

To cut  $x$ -axis

$$\sin 5x + \sin x = 0$$

$$\sin(3x+2x) + \sin(3x-x) = 0$$

ie)  $2\sin 3x \cos 2x = 0$  ✓

So  $\sin 3x = 0$  or  $\cos 2x = 0$

$$3x = 0, \pi, 2\pi, 3\pi, \dots$$

or  $2x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}$  ✓

∴  $x = 0, \frac{\pi}{3}$  or  $x = \frac{\pi}{4}$  ✓

(b) For  $y = \tan^{-1}(\tan x)$

(i) Domain: all real  $x$ 

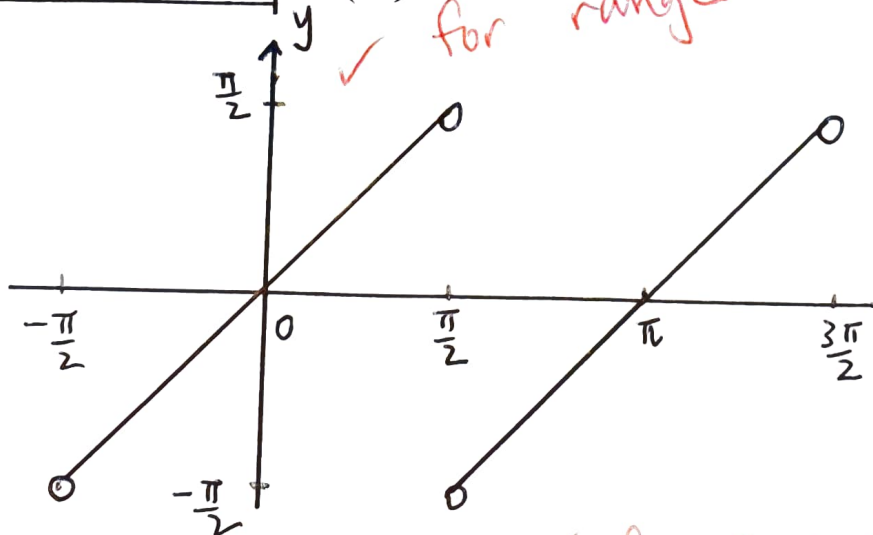
except  $x = \frac{(2n+1)\pi}{2}$  ✓

where  $n$  is an integer

&amp; Odd function ✓

(ii)  $\tan^{-1} \tan x = x$  ✓

(iii)



✓ for range

✓ for 2 straight line

□ Carried error for 'open circle' from (i)